

MATERIAL SAFETY DATA SHEET

1. PRODUCT AND MANUFACTURER IDENTIFICATION

FIPRONIL 20% W/V

SUPPLIER NAME:	Zhejiang Funong Biotech.Co., Ltd..
SUPPLIER ADDRESS:	Lantian Industrial Zone,Haibin Sub-District, Longwan District, Wenzhou, Zhejiang, China
SUPPLIER PHONE:	+86 577-88957117
SUPPLIER FAX:	+86 577-86875158
CHEMICAL NAME	(±)-5-amino-1-(2,6-dichloro- α,α,α -trifluoro-p-tolyl)-4-trifluoromethylsulfinylpyrazole-3-carbonitrile
CAS NUMBER	120068-37-3
MOLECULAR FORMULA	C ₁₂ H ₄ Cl ₂ F ₆ N ₄ O ₂ S
MOLECULAR WEIGHT	437.15
CHEMICAL FAMILY	phenylpyrazole
USE	Insecticides

2.COMPOSITION / INFORMATION ON INGREDIENTS

Fipronil	20% W/V
Other ingredients	up to 1L

3.HAZARDS IDENTIFICATION

2.1Classification of the substance or mixture

Acute toxicity - Oral, Category 3

Acute toxicity - Dermal, Category 3

Acute toxicity - Inhalation, Category 3

Specific target organ toxicity – repeated exposure, Category 1

Hazardous to the aquatic environment, short-term (Acute) - Category Acute 1

Hazardous to the aquatic environment, long-term (Chronic) - Category Chronic 1

2.2GHS label elements, including precautionary statements



Pictogram(s)

Signal word

Danger

Hazard statement(s)

H301 Toxic if swallowed

H311 Toxic in contact with skin

H331 Toxic if inhaled

H372 Causes damage to organs through prolonged or repeated exposure

H410 Very toxic to aquatic life with long lasting effects

Precautionary statement(s)

Prevention

P264 Wash ... thoroughly after handling.

P270 Do not eat, drink or smoke when using this product.

P280 Wear protective gloves/protective clothing/eye protection/face protection.

P261 Avoid breathing dust/fume/gas/mist/vapours/spray.

P271 Use only outdoors or in a well-ventilated area.

P260 Do not breathe dust/fume/gas/mist/vapours/spray.

P273 Avoid release to the environment.

Response

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P301+P310 IF SWALLOWED: Immediately call a POISON CENTER/doctor/...

P321 Specific treatment (see ... on this label).

P330 Rinse mouth.

P302+P352 IF ON SKIN: Wash with plenty of water/...

P312 Call a POISON CENTER/doctor/...if you feel unwell.

P361+P364 Take off immediately all contaminated clothing and wash it before reuse.

P304+P340 IF INHALED: Remove person to fresh air and keep comfortable for breathing.

P311 Call a POISON CENTER/doctor/...

P314 Get medical advice/attention if you feel unwell.

P391 Collect spillage.

Storage

P405 Store locked up.

P403+P233 Store in a well-ventilated place. Keep container tightly closed.

Disposal

P501 Dispose of contents/container to ...

4. FIRST - AID MEASURES

Eye Contact: Hold eyes open and rinse slowly and gently with water for 15-20 minutes. Remove contact lenses, if present, after the first 5 minutes, then continue rinsing eyes. Call a poison control center or doctor for treatment advice.

Skin Contact: Take off contaminated clothing. Rinse skin immediately with plenty of water for 15-20 minutes. Call a poison control center or doctor for treatment advice.

Inhalation: Move person to fresh air. If person is not breathing, call an emergency responder or ambulance, then give artificial respiration; if by mouth to mouth use rescuer protection (pocket mask etc). Call a poison control center or doctor for treatment advice.

Ingestion: No emergency medical treatment necessary.

Notes to Physician: No specific antidote. Treatment of exposure should be directed at the control of symptoms and the clinical condition of the patient. Have the Safety Data Sheet, and if available, the product container or label with you when calling a poison control center or doctor, or going for treatment.

5. FIRE FIGHTING MEASURES

Extinguishing Media: To extinguish combustible residues of this product use water fog, carbon dioxide, dry chemical or foam.

Fire Fighting Procedures: Keep people away. Isolate fire and deny unnecessary entry. To extinguish combustible residues of this product use water fog, carbon dioxide, dry chemical or foam. Contain fire water run-off if possible. Fire water run-off, if not contained, may cause environmental damage.

Special Protective Equipment

for Firefighters: Wear positive-pressure self-contained breathing apparatus (SCBA) and protective firefighting clothing (includes firefighting helmet, coat, trousers, oots, and gloves). If protective equipment is not available or not used, fight fire from a protected location or safe distance.

Unusual Fire and Explosion Hazards: This material will not burn until the water has evaporated. Residue can burn. If exposed to fire from another source and water is evaporated, exposure to high temperatures may cause toxic fumes.

Hazardous Combustion Products: Under fire conditions some components of this product may decompose. The smoke may contain unidentified toxic and/or irritating compounds. Combustion products may include and are not limited to: Nitrogen oxides. Carbon monoxide. Carbon dioxide.

6. ACCIDENTAL RELEASE MEASURES

Steps to be Taken if Material is Released or Spilled: Contain spilled material if possible. Small spills: Absorb with materials such as: Clay. Dirt. Sand. Sweep up. Collect in suitable and properly labeled containers. Large spills: Contact Dow AgroSciences for clean-up assistance.

Personal Precautions: Use appropriate safety equipment. For additional information, refer to Section 8, Exposure Controls and Personal Protection.

Environmental Precautions: Prevent from entering into soil, ditches, sewers, waterways and/or groundwater. See Section 12, Ecological Information.

7. HANDLING AND STORAGE

General Handling:

Keep out of reach of children. Do not swallow. Avoid breathing vapor or mist. Avoid contact with eyes, skin, and clothing. Use with adequate ventilation. Wash

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thoroughly after handling.

Storage

Store in a dry place. Store in original container. Keep container tightly closed when not in use. Do not store near food, foodstuffs, drugs or potable water supplies.

8. EXPOSURE CONTROLS / PERSONAL PROTECTION

Respirator: An approved respirator suitable for protection from dusts and mists of pesticides is adequate. Limitations of respirator use specified by the approving agency and the manufacturer must be observed.

Clothing: Employee must wear appropriate protective (impervious) clothing and equipment to prevent repeated or prolonged skin contact with this substance.

Gloves: Employee must wear appropriate synthetic protective gloves to prevent contact with this substance.

Eye Protection: The use of goggles is recommended. Emergency eye wash: Where there is any possibility that an employee's eyes may be exposed to this substance; the employer should provide an eye wash fountain or appropriate alternative within the immediate work area for emergency use.

9. PHYSICAL AND CHEMICAL PROPERTIES

APPEARANCE FORM:	White to Off-White liquid
ODOUR	Characteristic
ODOUR THRESHOLD	No data available
PH	6.41 (1% solution)
MELTING POINT/FREEZING POINT	200°C
INITIAL BOILING POINT AND BOILING RANGE	Decomposes before boiling
FLASH POINT	No data available
EVAPORATION RATE	No data available
FLAMMABILITY (SOLID, GAS)	No data available
UPPER EXPLOSION LIMIT:	No data available
LOWER EXPLOSION LIMIT:	No data available
VAPOUR PRESSURE	3.7×10 ⁻⁷ Pa at 25°C
VAPOUR DENSITY	No data available
RELATIVE DENSITY	1.08g/cm
WATER SOLUBILITY	Soluble
PARTITION COEFFICIENT: NOCTANOL/WATER	Log P 6.6
SURFACE TENSION	No data available
HENRY'S LAW CONSTANT	No data available
AUTO-IGNITION TEMPERATURE	No highly flammable
DECOMPOSITION TEMPERATURE	No data available
VISCOSITY	No data available
EXPLOSIVE PROPERTIES	No explosive properties
OXIDIZING PROPERTIES	No oxidizing properties

10. STABILITY AND REACTIVITY

Stability/Instability: Thermally stable at recommended temperatures and pressures.

Conditions to Avoid: Active ingredient decomposes at elevated temperatures.

Incompatible Materials: None known.

Hazardous Polymerization: Will not occur.

Thermal Decomposition: Decomposition products depend upon temperature, air supply and the presence of other materials. Decomposition products can include and are not limited to: Carbon monoxide. Carbon dioxide. Nitrogen oxides.

11. TOXICOLOGICAL INFORMATION

ACUTE TOXICITY:

ORAL LD50 (RAT): 300-2000 mg/Kg

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DERMAL LD50 (RABBIT):	>5000 mg/kg
INHALATION (RAT):	> 2.18 mg/L
EYE IRRITATION (RABBIT):	No eye irritation
SKIN IRRITATION (RABBIT):	Not a dermal irritant
SKIN SENSITIZATION (RAT):	Not a dermal sensitizer

12. ECOLOGICAL INFORMATION

Mobility, Degradability & Accumulation:

In plants, animals and the environment, Fipronil is metabolised via reduction to the sulphide, oxidation to the sulfone, and hydrolysis to the amide. In the presence of sunlight, a photodegradate also forms via sulfoxide extrusion. The sulphide, sulfone and photodegradate are known to act at the GABA receptor site, whereas the amide does not. Animals In rats, once absorbed, the distribution and metabolism of Fipronil is rapid.

Elimination is mainly via the faeces as Fipronil and its sulfone. The two major urinary metabolites were identified as conjugates of ring-opened pyrazole products. The distribution of radioactive residues in tissues was extensive after seven days. In goats and hens, the sulfone was the only metabolite identified in tissues.

Plants When applied as an incorporated soil treatment to cotton, maize, sugar beet or sunflowers, uptake of Fipronil into plants in all cases was low (c. 5%). At crop maturity, the major residue components observed in all plants were Fipronil, the sulfone, and the amide. Following foliar application to cotton, cabbage, rice and potatoes, at crop maturity, Fipronil and the photodegradate were the major residue components.

ECOTOXICOLOGY:

Bees: Highly toxic to honeybees, both by direct contact and by ingestion. However, no risk to bees when used as a soil or seed treatment.

Birds: Acute oral LD50 for bobwhite quail 11.3, mallard ducks >2000, pheasants 31, red-legged partridges 34, house sparrows 1120, pigeons >2000 mg/kg.

Dietary LC50 (5 d) for bobwhite quail 49, mallard ducks >5000 mg/kg diet.

Fish: Acute LC50 (96 h) for bluegill sunfish 85, rainbow trout 248, European carp 430 µg/l.

Daphnia: LC50 (48 h) 0.19 mg/l; for *D. carinata* (48 h) 3.8mg/l.

Algae: EC50 (96 h) for *Scenedesmus subspicatus* 0.068 mg/l; (120 h) for *Selenastrum capricornutum* >0.16, *Anabaena flos-aquae* >0.17 mg/l.

Earthworms: Non-toxic.

13. DISPOSAL CONSIDERATIONS

Product Disposal: On site disposal of the concentrated product is not acceptable. Ideally, the product should be used for its intended purpose. If there is a need to dispose of the product, approach local authorities who hold periodic collections of unwanted chemicals.

Container Disposal Do not use this container for any other purpose.

Triple rinse containers, add rinsate to the spray tank, then offer the container for recycling/reconditioning, or puncture top, sides and bottom and dispose of in landfill in accordance with local regulations.

If recycling, replace cap and return clean containers to recycler or designated collection point.

If not recycling, puncture or shred and bury containers in local authority landfill.

If no landfill is available, bury the containers below 500mm in a disposal pit specifically marked and set up for this purpose clear of waterways, desirable vegetation and tree roots.

Empty containers and product should not be burnt.

14. TRANSPORT INFORMATION

Transport: Considered non dangerous for road and rail transport (in packaging, including IBCs not greater than 3000 Lt each) by the Australian Code for the Transport of Dangerous Goods by Road and Rail. Ref: ADG7; SP No. AU01.

For bulk shipments as: Class 9, use UN 3082, PGIII, HazChem code 2Z.

UN Number (Sea Transport): 3082

IMO Class/Packing Group: Class 9/PGIII

15. REGULATORY INFORMATION

Safety, Health and Environmental Regulations/Legislation Specific for The Substance or Mixture

This safety datasheet complies with the requirements of Regulation (EC) No. 1907/2006.

16. OTHER INFORMATION

The data given here is based on current knowledge and experience. The purpose of this safety data sheet is to describe the products in terms of their safety

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requirements. The data does not signify any warranty with regard to the products' properties.